

DENTAL MATERIALS

Chapter 3

Mechanical Properties

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Mechanical properties

They are properties that describe the reaction of the material under force or load.

- They are important because restorative materials must withstand forces during service (fabrication & mastication).
- They are determined by the nature and strength of atomic bond.

Force:

- **Def.:** It is action which produces or tends to produce a change in the state of rest or motion of a body

- **Force may cause** → Displacement.

→ Acceleration.

→ Deformation: if the body remains at rest.

- **Force is characterized by:**

a. Speed → determine whether the force is static or dynamic.

b. Magnitude.

c. Point of application: long axis or away

d. Direction: axial or tangential

- **Unit:** Newton (N) or Pound (lb) $1\text{ lb} = 4.4\text{ N}$

❖ Effect of force on the body: the body is in a state of equilibrium:

a) External equilibrium: 2 forces opposite in magnitude and equal in direction.

b) Internal equilibrium: the body being in integral shape where internal reaction balances the external applied forces (stress)

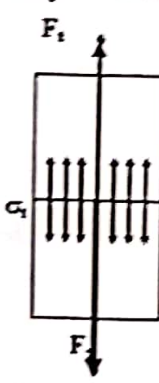
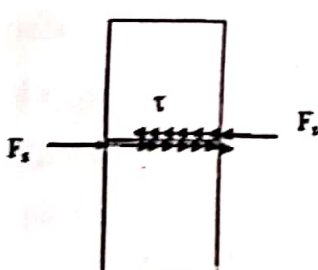
1) Stress (σ):

- **Def.:** It is the internal reaction to the external applied force.
- **Equation:** $\text{Stress} = \frac{\text{force}(F)}{\text{area}(A)}$
- **Unit:** - Pa = N/m² - MPa = MN/m² = N/mm² - Kg/cm² - lb/in²

➤ **N.B:** It is equal in magnitude and opposite in direction to the applied force.

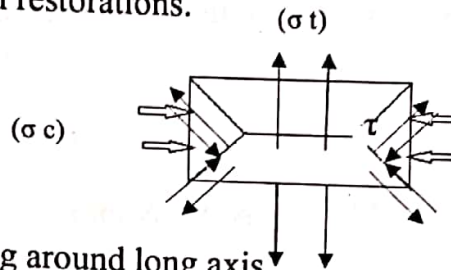
- Types of stresses: the molecules of the body resist the forces

(الضغط على شكل واحد)

Normal (axial) stresses		Tangential stresses
A) Tensile stress (σ_t)	B) Compressive stress (σ_c)	C) Shear stress (τ)
- Result when body is subjected to: <u>2 sets of force.</u> i. Directed away from each other. ii. On the same straight line. - Body → elongation.	- Result when body is subjected to: <u>2 sets of force.</u> i. Directed toward each other. ii. On the same straight line. - Body → shortening.	- Result when body is subjected to: <u>2 sets of force.</u> i. Directed toward each other. ii. Not on the same straight line. - Body → tearing or sliding.
		

D) Complex stresses:

Results when all types of stress develop in a structure as a result of application of load e.g. most dental restorations.



E) **Torsion** : twisting around long axis

F) **Bending**: stress away from fixed point

2) Strain (ε):

- **Def.:** Change in length (deformation) per unit length when the material is subjected to force.

➤ *N.B: the deformation that results due to stress produced within the material*

- **Equation:** $\text{Strain } (\epsilon) = \frac{\text{deformation } (\Delta L)}{L_o} = \frac{L_f - L_o}{L_o}$

- **Unit:** no unit.

- **Types of strains:**

A) **Elastic strain** (reversible or temporary): the material will recover to its original shape on removal of the external force.

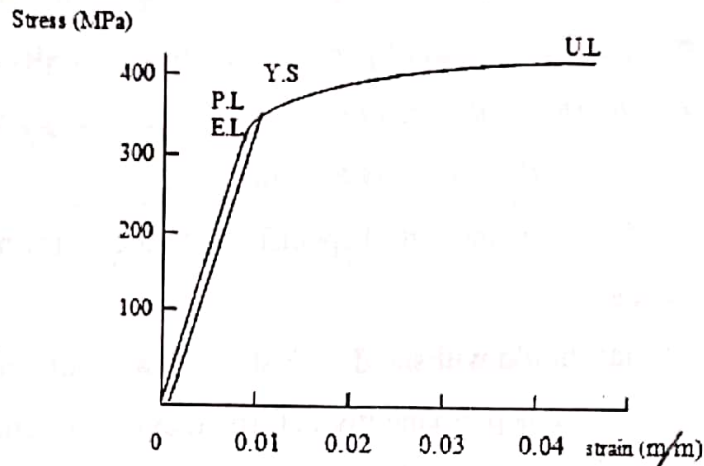
B) **Plastic strain** (irreversible or permanent): the material will not recover to its original shape on removal of external force.

➤ **Poisson's ratio (μ):**

- **Def.:** It is the ratio of lateral strain to axial strain within elastic limit.
- It indicates that the reduction in cross section is proportional to elongation during elastic strain.

$$\text{Poisson's ratio } (\mu) = \frac{\text{lateral strain}}{\text{axial strain}} \quad \text{usually } 0.3$$

- **Relation between stress and strain (stress-strain curve):**



- The engineering stress-strain curve is that curve where the applied load is divided by the original cross section to calculate the strain.

- Hook's law: strain is directly proportional to the stress until certain limit known as proportional limit.

1) **Proportional limit (P.L):**

- **Def.:** It is the maximum stress the material can withstand without deviation from hook's law.
- It describes the relation between stress and strain.

2) **Elastic limit (E.L):**

- **Def.:** It is the maximum stress the material can withstand without permanent deformation.
- It describes the elastic behavior of the material.

➤ *N.B: The proportional limit and the elastic limit represent the same stress value.*

3) Yield strength (Y.S) :

- **Def.:** It is the stress at which the material begins to function in a plastic manner (specific amount of permeant deformation occur).
- **N.B:** It is the stress at which the material shows a specific limiting deviation from proportionality of stress to strain.
- **Percent offset:** the amount of specific permanent strain (0.1, 0.2, and 0.5%).
- **Importance:**
 - a. The material should withstand high stresses without permanent deformation.
 - b. If the restoration is permanently deformed even without breakage it is considered a functional failure.
 - c. To adjust a restoration, we must apply stress greater than yield to produce permanent deformation e.g. burnishing.

4) Ultimate strength:

- **Def.:** It is the maximum stress the material can withstand before fracture.
- **N.B:** The yield stress is more important than ultimate stress, because yield stress represents the clinical failure (functional failure) where the material will start to permanently deform.

5) Modulus of elasticity (Young's Modulus) 'E':

(Elastic defor) = $\frac{\Delta l}{l_0}$

- **Def.:** It is the constant of proportionality between stress and strain in axial (normal) stress within elastic limit.
- It measures the rigidity or stiffness.
- **N.B:** It represents the slope of elastic portion of stress-strain curve.

- **Equation:** Modulus of elasticity (E) = $\frac{\text{stress}(\sigma)}{\text{strain}(\epsilon)}$

- **Unit:** -MPa = MN/m^2 - Kg/cm^2 - lb/in^2

➤ *N.B: Modulus of elasticity doesn't change if tested under compression or tension as it depends on interatomic & intermolecular forces but it doesn't depend on heat treatment or mechanical treatment.*

➤ *Metals & alloys > plastic and rubbers*

- **Importance:**

Materials with high modulus of elasticity have an even stress distribution, so it can be used in thin sections.

a. Long span bridges → to allow better stress distribution.

b. Denture base materials → to be used in thin sections without the fear of bending and uneven stress distribution.

➤ *N.B: (E) of cobalt chromium > (E) of gold alloys.*

6) Modulus of shear (G)

- **Def.:** It is the constant of proportionality between stress and strain in ^{ratio} non-axial (tangential) stress.

- **Equation:** Modulus of shear (G) = $\frac{\text{modulus of elasticity}}{2(1+\text{poisson's ratio})} = \frac{E}{2(1+\mu)}$

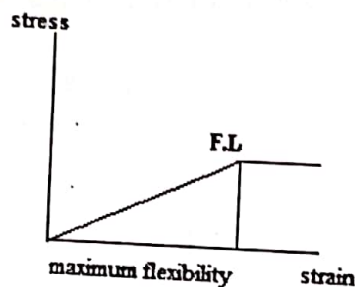
- *N.B: G is about 40 % of E*

* Base under Amalgam should be rigid → to support brittle

* $\text{///} \text{ } \text{//} \text{ Porcelain } \text{///} \xrightarrow{\text{Amalgam}} \text{///} \text{ } \text{//} \text{ Porcelain } \text{///}$

7) **Flexibility:**

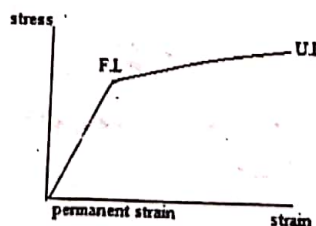
- **Def.:** the strain resulting in the material when the stress reaches the elastic limit (stressed to proportion limit).
- **Equation:** maximum flexibility(ϵ_m) = $\frac{\text{proportional limit(P)}}{\text{Modulus of elasticity(E)}}$



- **Importance:** Orthodontic wires, endodontic files and clasps of partial denture and elastic impression materials.

8) **Ductility and malleability**

- It is a property of metals and alloys
- **Ductility:** it is the ability of a material to be plastically deformed under tension without fracture.
- e.g.: Material is drawn into wires.
- **Malleability:** it is the ability of a material to plastically deform under compressive without fracture.
- e.g.: Material is hammered or rolled into thin sheets.

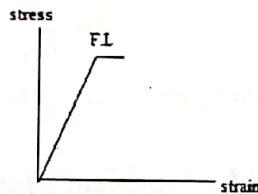


➤ *N.B: ductile materials fracture by necking (decrease in the area of fracture)*

- **Importance:** indicates the workability of the materials as the material when being burnished.

9) **Brittleness:**

- Brittle material fracture at or near proportion limit.
- Brittle material shows NO or very little plastic deformation.
- Brittle materials are weak in tension.



- *Amalgam has compressive strength 6 times higher than tensile strength.*
- *Brittle materials fracture by crack propagation.*
- *Brittle materials may have high strength.*

10) **%Elongation:**

- **Def.:** It is the deformation results from the application of tensile strength.

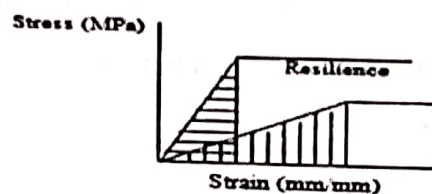
- **Equation:**
$$\text{Elongation (\%E)} = \frac{\text{increase in length}}{\text{original length}} \times 100$$

➤ *N.B:*

- *Ductile alloys (gold) show more 20% elongation while brittle (nickel-chromium alloy) shows 1% of elongation.*
- *This means that gold elongates 1/5 of its length before fracture.*

11) Resilience:

- **Def.:** It is the amount of energy absorbed when the material is stressed to its proportional limit.
- It is measured by the area under the elastic portion in the stress-strain curve.
- **Equation:** Resilience (R) = $\frac{1}{2} \text{ stress} \times \text{strain} = \frac{1}{2} \sigma \cdot \epsilon$
- **Unit:** mMN/m³



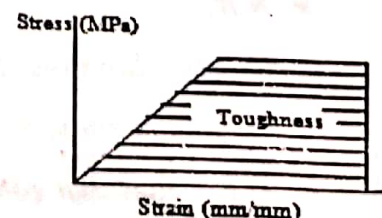
• **Importance:**

- Orthodontic wires should be resilient as the stored energy is released causing movement of teeth.
- Denture base should be resilient to absorb masticatory forces and transmit less force to underlying bone.

12) Toughness:

(بالتحمل مع الحد الأقصى)

- **Def.:** Amount of energy absorbed when the material is stressed to the point of fracture.
- It is represented by the area under elastic and plastic portion of the stress-strain curve.
- **Unit:** mMN/m³
- **N.B:** materials of high strength and high ductility have high toughness.



13) Fracture toughness:

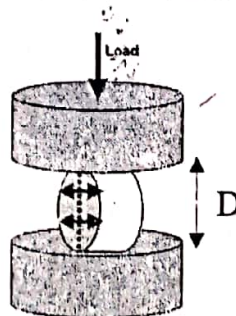
(يستخدم مع الكوارس الخشنة defect)

- **Def.:** It is the amount of energy required to fracture material with defects (flaws or cracks)
- **N.B:** Brittle materials have low fracture toughness, while ductile materials have high fracture toughness.
- **Importance:**
 - a) It gives a relative value of a material's ability to resist crack propagation under tensile stresses
 - b) Materials should be designed to resist this type of failure.

• Other mechanical properties:

1. Diametral compression test (Indirect tensile test or Brazilian test):

- Test to measure the tensile strength of a brittle material by applying compressive load.
- The compressive stress induces a tensile stress in a material.
- **Equation:** Tensile stress (σ_t) = $\frac{2 \times \text{load}}{\pi \times \text{diameter} \times \text{thickness}} = \frac{2P}{\pi DT}$

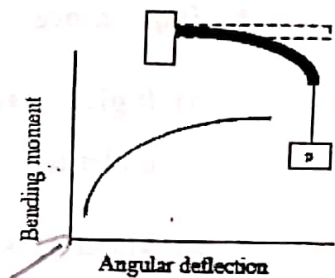


2. Cantilever bending:

- It is measured by clamping a sample at one point and applying a force at fixed distance from the other side.
- As the force is increased the angle of bending and bending moment (force $\times$ distance) is increased.
- *N.B: If a curve is drawn between the bending moment and angle of bending it will be similar to stress strain curve.*

• Importance:

The bending properties of stainless steel wires, endodontic files and reamer and needles are important.



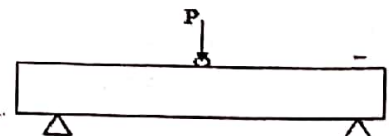
بجانب الـ P من بعد الـ 20cm - 20cm = 40cm

3. Transverse strength (flexure strength or modulus of rupture):

• Three point bending test:

- It is obtained by subjecting a bar fixed from both sides to a load applied on the middle.
- The test determines the strength of the material (σ) and the amount of deformation expected.

• Equation:



$$\text{Stress } (\sigma) = \frac{3 \times \text{load} \times \text{length}}{2 \times \text{width} \times \text{thickness}^2} = \frac{3PL}{2bd^2}$$

قلب هذا

$$\text{Deformation} = \frac{\text{load} \times \text{length}^3}{4 \times \text{Elastic modulus} \times \text{width} \times \text{thickness}^3} = \frac{PL^3}{4Ebd^3}$$

(depth)

بلغ ابعث 4

$$\begin{array}{l} \text{Stress} \quad \frac{3Pl}{2bd^2} \\ \text{Load} \quad \frac{Pl^3}{4Ed^3} \\ \text{Def.} \quad \frac{Pl^3}{4Ed^3} \end{array}$$

width $\rightarrow$ Thickness

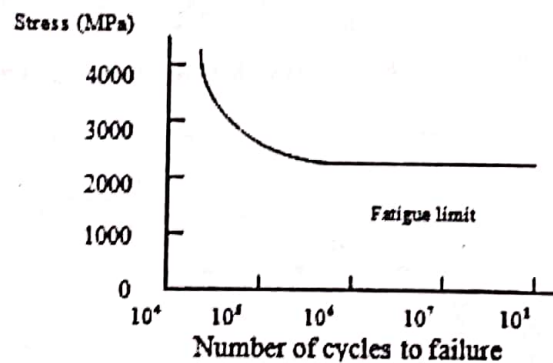
- **Importance:**

a. Denture base are subjected to this type of stresses during mastication.

b. In long span bridges where deformation $\propto \frac{\text{length}^3}{\text{width}^3}$
 thickness (depth)

4. Fatigue:

- **Def.:** fracture of the material when subjected to repeated (cyclic) loading below yield strength.



- **Fatigue limit:** number of cycles performed until fracture occurs.

- **Endurance limit:** is the stress that can be applied an infinite number of cycling loading without causing material failure.

- **Importance:**

Any dental restoration should be designed so that the fatigue stresses under repeated masticatory loading should be less than fatigue limit

5. ^{مدرسة} Impact strength:

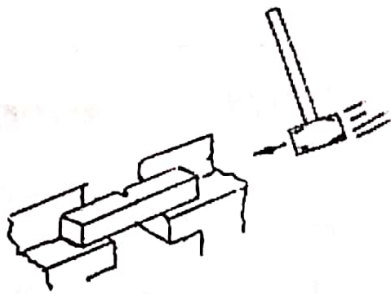
- **Def.:** It is the amount of energy required to fracture the material under sudden (impact) force.

e.g.: dropping of restorations.

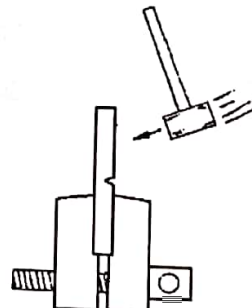
- It is measured by clamping a specimen, and then struck with pendulum so that the energy absorbed can be measured.

- **Tests:**

- a. Charpy test: the specimen is supported at both ends and struck at the middle.
- b. Izod test: the specimen is supported at one end and struck at the other end.



Charpy test



Izod test

- Unit: joules.

- **Importance:**

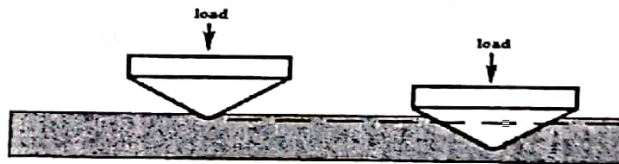
It predicts behavior of materials under sudden conditions

➤ *N.B: glass, amalgam and cements have low impact strength.*

6. Hardness:

- **Def.:** It is the resistance of the material to permanent indentation, penetration or scratching.
- It depends on the penetration of indenter into a surface area of the material where the smaller the indentation the higher the hardness and vice versa.

➤ *N.B: It is a surface property that cannot be obtained from stress strain curve.*



Test method	Indenter	Measurements	Notes
Brinell (BHN)	Steel ball	Area of indentation	➤ Not used to measure hardness of <u>-brittle materials</u> → cause fracture <u>-elastic materials</u> → cause elastic deformation not indentation ➤ It is used for metals and alloys
Rockwell	Steel ball or diamond point	Depth of indentation	➤ Used for testing brittle materials ➤ Its modification can be used for elastic materials
Vicker (VHN)	Diamond	Area of indentation	➤ Used to measure hardness of very hard materials <i>e.g.: brittle materials, tooth structure and gold alloys</i>
Knoop (KHN) Micro-indentation test	Diamond	Area of indentation	<u>Adv.</u> → used to measure brittle and elastic materials. <u>Disadv.</u> → specimen must be highly polished and needs long time
Shore A	Blunt point	Depth of penetration	➤ Used to measure hardness of rubbers

universal hardness test

- **Importance:**

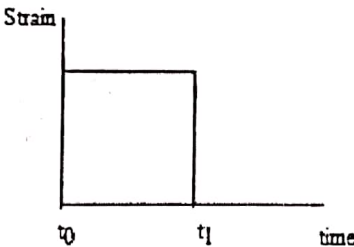
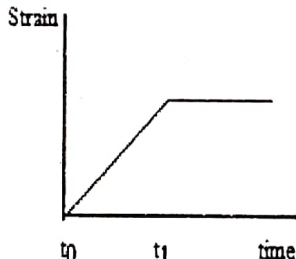
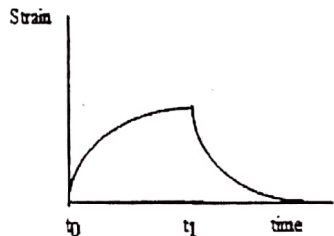
- a. Natural teeth should not be opposed by harder materials like porcelain.
- b. Avoid scratching of soft materials (model and die materials) because it decrease the accuracy.
- c. Hard restoration (Cobalt chromium):
 - Very difficult to finish and polish.
 - Once it is polished it maintain polished surface without scratching.

7. **Wear:**

- It is the loss of material resulting from mechanical action.
- *N.B: removal process occurs when surfaces slide over each other's.*
- Wear of teeth and restoration may result from:
 - a. Mechanical (abrasion) → brushing of teeth.
 - b. Physiological → attrition during normal mastication.
 - c. Pathological → bruxism.
- Wear is useful during finishing and polishing of restoration.

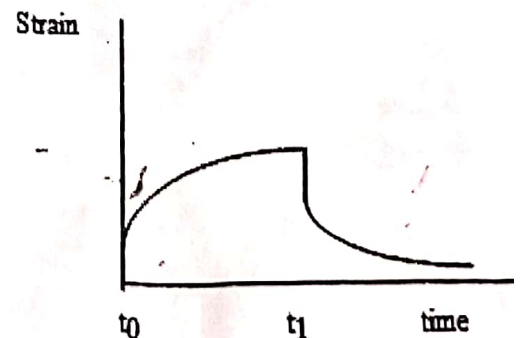
8. **Viscoelasticity:**

The rate of loading is important when dealing with strain rate sensitive materials like agar, alginate, rubber impression, amalgam, waxes, dentin, oral mucosa and periodontal ligaments.

A. Ideal elastic materials	B. Ideal viscous materials	C. Anelastic materials
- <u>On application of load</u> The strain remains constant with time - <u>On removal of load</u> The strain decreases to zero (immediate recovery) ➤ N.B: strain doesn't depend on rate of loading or time of load application 	- <u>On application of load</u> The strain uniformly increases with time - <u>On removal of load</u> The strain remains (no recovery) ➤ N.B: strain is directly proportion to the time. 	- <u>On application of load</u> Non-linear increase of strain with time. - <u>On removal of load</u> Non-linear recovery with time till zero 

D. Viscoelastic behavior:

- Viscoelastic materials are strain-rate sensitive materials
 - It is a combination of elastic, anelastic and viscous material.
- On application of load
- Immediate strain of the elastic part
 - Gradual increase of strain with time due to viscous and anelastic part.
- On removal of load
- Immediate recovery of elastic part.
 - Non-linear recovery of the anelastic part.
 - No recovery for the viscous part.



- **Importance:**

Elastic impression materials are viscoelastic which must be removed rapidly from the mouth with snap removal (high rate of loading) in order to:

- Decrease permanent deformation which results due to viscous part.
- Increase the strength (less chance to tear)
- They should be given a time after removal from mouth before pouring to give time for the anelastic part to recover.
- *N.B: compressive stress of viscoelastic materials increases with the increase of the rate of loading, thus strength of the materials should be tested under the same rate of loading.*

9. **Creep:**

- **Def.:** - Time dependant permanent deformation.

- At stress below yield strength.
- At temperatures near softening point.

- **Importance:**

- a. The softening temperature of most metals and ceramics are so far from the room or mouth temperature, they do not creep in dental application.
- b. In dental amalgam, they contain components have melting temperature near room temperature, so they may undergo creep.
- c. Many polymers as wax and rubbers have softening temperature near room temperature, so they undergo creep.

➤ *N.B: creep is termed flow for amorphous materials such as wax.*